PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS

Glossary of terms

Lectures 1–7

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Each term is given as it is used in the lectures. The label on the right gives the lecture in which the term is introduced (L2 = Lecture 2), followed by the lectures that use it again. Section 2 lists the symbols.

Terms

A

Adaptive method L2

A time-stepping method that estimates the error at each step and adjusts the step h automatically, e.g. RK45. *Adaptive quadrature* (L4) does the same for an integral: it splits the interval, estimates the error on each part, refines only where needed, and returns the integral with an error estimate.

Algorithmic complexity L1, L6

How the cost of a computation grows with the size n of the problem. $O(n)$: cost roughly proportional to n, so doubling n doubles the work; $O(n^2)$: doubling n gives four times the work. The FFT needs about $N \log_2 N$ operations instead of the N^2 of the direct DFT.

Aliasing L6

A component at a frequency above the Nyquist frequency f_N appears in the sampled data at a wrong, lower frequency $|f - m f_s|$, with the integer m that brings it below f_N. It cannot be undone after sampling.

Amplitude L6

A in $x(t) = A\cos(2\pi f t + \varphi)$: the largest deviation of the oscillation from zero. From the DFT, $A_k = 2|X_k|/N$ for $0 < k < N/2$; with a window, divide by $\sum w_n$ instead of N.

Amplitude spectrum L6

The amplitudes A_k plotted against the frequencies f_k: one value every $\Delta f = 1/T$.

Angular frequency L6

$\omega = 2\pi f$ (rad s^{-1}): the phase advance per second.

Association L7

In statistics texts, used much as dependence (*see* Dependence). Correlation coefficients are often called measures of association.

Asymmetric uncertainty L5

An uncertainty that is larger on one side of the value than on the other, as when a curved function is applied to an uncertain input. Quoted as an interval, e.g. the middle 68% of Monte Carlo results.

Attractor L7

The set of states that trajectories approach in the long run. A stable fixed point is a *point attractor*; an attracting limit cycle is a *cycle attractor* (e.g. van der Pol: from any initial condition except the fixed point at the origin, the trajectory ends on its limit cycle).

Autocorrelation L6, L7 (from L5)

The correlation of a series with itself shifted by k samples:

$$r_k = \frac{\sum_{n=0}^{N-1-k}(x_n - \bar{x})(x_{n+k} - \bar{x})}{\sum_{n=0}^{N-1}(x_n - \bar{x})^2}.$$

It shows how long a value influences those that follow. r_1 is the lag-1 autocorrelation of L5.

Autonomous system L2

A dynamic system whose rule does not change in time: shifting the time origin changes nothing. *see* Nonautonomous system.

B

Basin of attraction L7

The set of initial conditions whose trajectories end on a given attractor. For the van der Pol limit cycle: the whole phase plane except the origin.

Bisection L3

Root finding for a continuous f with $f(a)f(b) < 0$: take the midpoint, keep the half in which the sign changes, repeat. Guaranteed to converge once a bracket is known, but slow: one binary digit per step, $n \approx \log_2[(b-a)/\varepsilon]$ steps for tolerance ε.

Bootstrap L5, L6

Estimates the uncertainty of a statistic (e.g. the median) without assuming any form of the distribution. Draw N values at random from the N measured values, each of which can be drawn again; compute the statistic; repeat B times. The standard deviation of the B results is the standard error of the statistic. Condition: independent values.

Brent's method L3

A root-finding hybrid: fast secant or interpolation steps, with a bisection step whenever a step leaves the bracket. Guaranteed like bisection, usually as fast as the secant method.

C

Catastrophic cancellation L1, L4

Loss of significant digits when two large, nearly equal numbers are subtracted: their small rounding errors can dominate the small difference. Examples: the sum-of-squares formula for the variance; a finite difference with too small an h.

Chebyshev points L4

Interpolation points that crowd towards the ends of the interval, $x_k = \cos\big((2k+1)\pi/(2n+2)\big)$, $k = 0, \ldots, n$. They suppress the oscillations of a high-degree polynomial near the ends (*see* Runge's example).

Chi-square L3, L5

The sum of squared residuals in units of their standard uncertainties, $\chi^2 = \sum_i (r_i/\sigma_i)^2$. *see* Reduced chi-square.

Closed system L2

Energy crosses the system boundary, mass does not (e.g. gas in a piston–cylinder). Check: the energy balance.

Condition number L3

κ: a measure of conditioning (L1), the factor by which relative errors in the input can grow in the result. Forming the normal equations squares it: $\kappa(A^\mathsf{T}A) = \kappa(A)^2$.

Conditioning L1

How sensitive a *problem* is to small changes in its input. Small input change, small output change: well-conditioned; small input change, large output change: ill-conditioned. A property of the mathematical problem itself, not of the numerical procedure.

Convergence L1, L2

A numerical procedure converges if refining the calculation (smaller step h, more points, more samples) brings the result closer to the exact answer: $y_h \to y$ as $h \to 0$.

Convergence check L2, L4

Repeat a calculation with h, $h/2$ (and $h/4$) and compare. For a method of order p the ratio of successive differences is about 2^p; no exact solution is needed.

Correlation matrix L7

The covariance matrix of standardised variables: correlations off the diagonal, 1 on it. PCA of standardised data is PCA of the correlation matrix.

Correlation (Pearson) L7

$r = s_{xy}/(s_x s_y)$, between -1 and 1, without units: how close the points lie to a straight line. It tells reliably whether two quantities rise together or one rises as the other falls. $r = 0$ does not mean unrelated (plot the pair): for any steady oscillation x and $\dot{x}$ are uncorrelated, since $x\dot{x} = \frac{1}{2}\,d(x^2)/dt$ averages to zero over whole periods. For independent measurements its usual p-value holds; for a time series it does not, because successive values are not independent: for slowly varying series that p-value is too small.

Correlation time L6

τ_c: the lag over which the autocorrelation falls by a factor e. For $x_n = ax_{n-1} + \varepsilon_n$, $r_k \approx a^k = e^{-\tau/\tau_c}$.

Coupling L7

A term in the equations: the state of one system enters the rule of another. It is a property of the dynamics; dependence is a property of the measured data. Coupling usually creates dependence, but a common driver creates dependence without coupling, and weak coupling in noise may leave dependence too small to see. Coupling has a direction; dependence and correlation do not, so correlation does not measure coupling.

Covariance L7

For N pairs (x_i, y_i), each measured on the same unit or at the same moment: $s_{xy} = \frac{1}{N-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. The variance extended to two variables, $s_{xx} = s_x^2$: positive when x and y tend to be on the same side of their means, negative when on opposite sides; units of x times units of y, so its size depends on the units. For m variables, the *covariance matrix* C, with the variances on its diagonal; with centred columns, $C = X^\mathsf{T}X/(N-1)$.

Covariance matrix of the parameters L3

For a linear fit, $C = (A^\mathsf{T}A)^{-1}$. Its diagonal gives the parameter uncertainties, $\sigma_{\theta_j} = \sqrt{C_{jj}}$; the off-diagonal terms give the correlations $\rho_{jk} = C_{jk}/\sqrt{C_{jj}C_{kk}}$.

CSTR L2, L3

Continuous stirred-tank reactor: a well-mixed vessel with continuous inflow and outflow, so that the outflow has the composition of the tank. The example of an open system.

Cubic spline L4

One cubic on each interval, joined so that value, slope and curvature are continuous. Error $O(h^4)$ for smooth f; the usual default interpolant for smooth data.

D

Data matrix L7

X, N rows $\times$ m columns: one row per observation (a unit, or a moment or window of a series), one column per variable. Centred, its SVD gives PCA; $C = X^\mathsf{T}X/(N-1)$ is the covariance matrix.

Degrees of freedom L3, L5

$\nu = N - p$: the number of values minus the number of parameters fitted to the same data. For the standard deviation one value is used up by $\bar{x}$, so $\nu = N - 1$.

Delay coordinates L7

A phase space reconstructed from one measured series: plot $x(t)$ against $x(t-\tau)$, and for more complex systems $x(t-2\tau), \ldots$. A common starting point for τ is the first zero of the autocorrelation; nearby τ should give much the same shape. For a deterministic system, enough delay coordinates reproduce the structure of the attractor (embedding theorem of Takens and of Mañé, 1981).

Dependence L7

Knowing one variable changes what we know about the other; independence: the distribution of one does not change when the other is known. Correlation detects only one tendency: both rising together, or one rising as the other falls (along a line for r, along any monotonic curve for r_S). Zero correlation does not exclude it: on the van der Pol limit cycle each x allows only two values of $\dot{x}$, yet $r = 0$: zero correlation, strong dependence. Dependence of any shape: *see* Mutual information.

Design matrix L3

For a model linear in its parameters, $f = \sum_j \theta_j \phi_j(x)$: $A_{ij} = \phi_j(x_i)/\sigma_i$, with data $b_i = y_i/\sigma_i$, so that $\chi^2 = \|A\boldsymbol{\theta} - \mathbf{b}\|^2$.

Deterministic L2

Given the state and the rule, the future is fixed. Repeated runs from the same initial conditions give the same result. *see* Stochastic.

Discrete Fourier transform (DFT) L6

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N}, \quad x_n = \frac{1}{N}\sum_{k=0}^{N-1} X_k e^{2\pi i k n/N}.$$

The complex X_k measures how much of the oscillation at $f_k = k/T$ the series contains: its modulus gives the amplitude, its argument the phase.

Discretisation error L2

The error caused by a finite step h. Halving h reduces it; it is distinct from round-off error and from physical uncertainty. It is the truncation error (L1, L4) of a time-stepping method.

Dynamic mode decomposition (DMD) L7

Fits linear dynamics $\mathbf{x}_{n+1} = A\mathbf{x}_n$ to snapshots of many variables; each eigenvector of A is a mode with its own frequency and growth or decay rate. Assumes linear, autonomous dynamics.

Dynamic system L2

A system whose state evolves in time; it needs time stepping. A *static* problem, $f(x) = 0$, needs root finding.

E

Eigenvalue and eigenvector L7

$A\mathbf{v} = \lambda\mathbf{v}$: a direction $\mathbf{v}$ that A only stretches, by λ. For $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(t) = \sum c_i e^{\lambda_i t}\mathbf{v}_i$: real part of λ negative – decay, positive – growth; imaginary part – oscillation. Near a fixed point, the eigenvalues of the Jacobian give the local motion.

Ensemble L2

Many runs of a stochastic model, or of a deterministic model with uncertain inputs. Each run is one *realisation*; the ensemble gives the mean, variance and distribution of the outcomes. The sampling error of the ensemble mean falls as $1/\sqrt{N}$.

Euler method L1, L2, L4

The simplest update rule, $y_{n+1} = y_n + h\,f(t_n, y_n)$, $t_{n+1} = t_n + h$: follow the tangent to the solution for one step. One slope evaluation per step, order 1 (RK1).

Euler–Maruyama method L2

Euler's method for stochastic dynamics $dX = f\,dt + g\,dW_t$: $X_{n+1} = X_n + f\,\Delta t + g\sqrt{\Delta t}\,\xi_n$, $\xi_n \sim \mathcal{N}(0,1)$. Strong order only $1/2$; the $h/2$ check does not carry over directly.

Explained variance L7

The fraction of the total variance along each principal direction, $\sigma_k^2 / \sum_j \sigma_j^2$. Keep the components above the noise floor.

Explicit and implicit methods L2

An explicit method computes the next value from known values only; for stiff problems it must take tiny steps to remain stable. An implicit method (backward Euler, BDF) solves an equation for the next value and remains stable with much larger steps.

Extrapolation L4

Estimating values outside the range of the data, where nothing in the data checks the result. The error of a polynomial or spline grows fast beyond the data; justified only with a physical model, and then stated as an assumption.

F

Fast Fourier transform (FFT) L6

An algorithm that computes exactly the same X_k as the DFT in about $N \log_2 N$ instead of N^2 operations (`numpy.fft.rfft`).

Finite differences L4

Derivatives from values at neighbouring points. Forward: $[f(x+h) - f(x)]/h$, order 1. Central: $[f(x+h) - f(x-h)]/(2h)$, order 2. Second derivative: $[f(x+h) - 2f(x) + f(x-h)]/h^2$, order 2. Too small an h makes round-off dominate.

Finite precision L1

A computer stores each number with a finite number of binary digits. A number such as $0.1_{10} = 0.000110011\ldots_2$, whose binary form is infinite, is therefore stored only approximately; hence `0.1 + 0.2` gives `0.30000000000000004`. The origin of round-off error.

Fit L3

Choosing the parameters $\boldsymbol{\theta}$ of a model $f(x;\boldsymbol{\theta})$ so that the model is as close as possible to the data; "closest" is defined by a chosen measure, usually least squares. A fit passes between the points; *see* Interpolation.

Fixed point L7

A state that does not change, $\dot{\mathbf{x}} = \mathbf{0}$: for the state $(x, \dot{x})$ both $\dot{x} = 0$ and $\ddot{x} = 0$. If all eigenvalues of the Jacobian there have negative real parts, it is an attractor (a point attractor); the van der Pol origin is not.

Frequency L6

f (Hz $= \mathrm{s}^{-1}$): oscillations per second; period $P = 1/f$. The DFT frequencies are $f_k = k/T$, $k = 0, \ldots, N/2$: the oscillations with k whole periods in the record, spacing $\Delta f = 1/T$.

Frozen system L7

The system with its time-varying parameters held at their current values. When the parameters change slowly, its fixed points and cycles move with them; when they change fast, it no longer describes the motion.

G

Gaussian distribution L5, from L3

The distribution that arises when many small independent effects add up; 68.3%, 95.4% and 99.7% of values lie within $\pm 1, 2, 3$ standard deviations of the mean. Assume it only when physics or an independent calibration supplies it.

Gauss–Legendre quadrature L4

For a function that can be evaluated anywhere: n points and weights chosen so that the rule is exact for polynomials up to degree $2n - 1$.

Gauss–Newton method L3

Newton's method for χ^2: replace the model by its tangent in the parameters, solve the resulting linear least-squares problem for the step, repeat. *Levenberg–Marquardt* takes shorter, more downhill steps far from the minimum.

Global error L2

The error accumulated at the end of a run, after all steps; $O(h^p)$ for a method of order p.

Golden-section search L3

A bracketing method for a minimum: keep three points with the lowest in the middle and shrink the interval by 0.618 each step. No derivatives; guaranteed for one minimum in the bracket.

Gradient descent L3

$x_{n+1} = x_n - \eta f'(x_n)$ (or $-\eta \nabla f$): step downhill. Simple; slow in narrow valleys. Like every local method, it finds *a* local minimum.

H

Hann window L6

$w_n = \sin^2(\pi n/N)$: multiplies the data so that they go smoothly to zero at both ends of the record, which reduces leakage. Cost: each peak is about two Δf wide.

Harmonics L6

Peaks at whole multiples of the fundamental frequency $f_0 = 1/P$. They describe the non-sinusoidal shape of *one* oscillation, not separate oscillations.

Hessian L3

$H_{ij} = \partial^2 f/\partial x_i \partial x_j$. Where $\nabla f = 0$, the point is a local minimum if all eigenvalues of H are positive (the many-dimensional $f'' > 0$).

Hodges–Lehmann estimate L5

The size of the difference between two groups: the median of all pairwise differences $B_j - A_i$, with an interval. It reports *how large* a difference is; a p-value reports only how surprising it would be under H_0.

I

Independent component analysis (ICA) L7

Unmixes signals that are fixed linear mixtures of statistically independent, non-Gaussian sources, by making the recovered sources as independent as possible.

Independent values L5, L6

Knowing one value tells nothing about the next. Standard errors, p-values and the bootstrap rely on it. (In L2, the four classification labels are also called independent: any combination can occur.)

Initial-value problem L1, L2

A starting point plus a rule for how it changes: $dy/dt = f(t, y)$, $y(t_0) = y_0$. Solved numerically by applying an update rule step by step.

Interpolation L4

Estimating values between the points by a simple function that passes *exactly* through every point. Suited to exact values (formulas, tables, simulation output); for measured values, fit first. Interpolation flattens fast oscillations and can shift phases.

Isolated system L2

Neither mass nor energy crosses the boundary (e.g. the two-body orbit). Check: the conserved quantities, energy and angular momentum.

J

Jacobian L3, L7

$J_{ij} = \partial F_i/\partial x_j$: replaces the derivative in Newton's method in several dimensions, $J\Delta\mathbf{x} = -\mathbf{F}$. In fitting, $J_{ij} = \partial f(x_i;\boldsymbol{\theta})/\partial\theta_j$. Near a fixed point $\mathbf{x}^*$ it linearises a system (L7): $\mathbf{f}(\mathbf{x}) \approx J(\mathbf{x} - \mathbf{x}^*)$, so $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$ with $\boldsymbol{\delta} = \mathbf{x} - \mathbf{x}^*$; its eigenvalues give the local motion, valid only close to $\mathbf{x}^*$.

L

Lagrange form L4

The unique polynomial of degree $\leq n$ through $n+1$ points: $p(x) = \sum_i f_i \ell_i(x)$, $\ell_i(x) = \prod_{j\neq i}(x - x_j)/(x_i - x_j)$.

Leakage L6

When the record does not hold a whole number of periods, the DFT, which treats the record as one period of a periodic signal, sees a jump where the end meets the start; the power of the oscillation spreads to all frequencies and its peak is lowered. Reduced by a window.

Least squares L3

Choose the parameters that minimise $S = \sum r_i^2$, or χ^2 when the points have uncertainties σ_i. For exact x_i and independent Gaussian errors this gives the maximum-likelihood parameters.

Likelihood L3

$p(d \mid \theta)$, the probability of the observed data as a function of the parameters: the model of the data. For counts, the product of Poisson probabilities; for Gaussian errors $-2\ln\mathcal{L} = \chi^2 + \text{const}$. Not a distribution of θ.

Limit cycle L2, L7

An isolated closed trajectory in phase space: a self-sustained oscillation. When attracting, trajectories from nearby initial conditions approach it – for van der Pol from every start except the fixed point at the origin; its size and shape are set by the system, not by the start. Possible only in nonlinear systems.

Linear and nonlinear dynamics L2, L3

A linear system obeys superposition: the response to a sum of inputs is the sum of the responses, and the frequency does not depend on the amplitude. A nonlinear system does not; it can show limit cycles, several steady states, chaos. This is a testable property of the system.

Linear in parameters L3

A model $f = \sum_j \theta_j \phi_j(x)$; least squares is then solved exactly, without iteration. A property of the fitting function, independent of whether the dynamics are linear.

Linear interpolation L4

The straight line between neighbouring points; error $\leq \frac{h^2}{8}\max|f''|$, order 2. Never overshoots.

Linear rule (propagation) L5

If f is nearly straight over $\bar{x} \pm \sigma_x$, $\sigma_y \approx |f'(\bar{x})|\sigma_x$; for independent inputs $\sigma_y^2 = \sum_i (\partial f/\partial x_i)^2 \sigma_i^2$ (adding in quadrature). When f is curved, use Monte Carlo.

Linearisation L3, L7

Replacing a function near a point by its tangent (in several variables: by the Jacobian). Newton's method uses it to find roots (L3); near a fixed point it turns $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ into $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$, whose eigenvalues give the local motion (L7). Valid only close to the point. *see* Jacobian.

Loadings L7

Name used in papers and software for the entries of a principal direction $\mathbf{v}_k$, the weights of the variables in that component; some software scales them by $\sigma_k/\sqrt{N-1}$. Check which is meant. (Not used in the slides.)

Local and global minimum L3

A local minimum is lowest in its neighbourhood; the global minimum is lowest overall. Local methods find a local minimum, which depends on the starting point.

M

MAD (median absolute deviation) L5

The median of the distances $|x_i - \text{median}|$. Half of the values lie within $\pm$MAD of the median, whatever the distribution; robust to bad values. Not to be confused with the *mean absolute deviation*, the mean of $|x_i - \bar{x}|$.

Maximum likelihood L3

The parameters that make the observed data most probable.

Mean L5

$\bar{x} = \frac{1}{N}\sum x_i$. Sensitive to a single bad value.

Median L5

The middle value when the values are sorted (for even N, the mean of the two middle ones). Robust to bad values.

Monte Carlo L2, L3, L5

Draw many values of the uncertain inputs, compute the result for each, and describe the distribution of results. Inputs: bootstrap samples of measured values, or draws from a stated tolerance. Works for any function; checks the linear rule.

Mutual information L7

How much knowing one variable reduces the uncertainty about the other; zero if and only if the two are independent, so it detects dependence of any shape. Mentioned for reference only.

N

Naive (sum-of-squares) variance formula L1

$\sigma^2 = \frac{1}{N}\sum x_i^2 - \bar{x}^2$. Mathematically identical to $\frac{1}{N}\sum(x_i - \bar{x})^2$, but numerically it subtracts two large, nearly equal numbers (catastrophic cancellation) and can give an inaccurate or even negative variance. *see* Welford's algorithm.

Newton–Raphson method L3

Replace f by its tangent and find where it crosses zero: $x_{n+1} = x_n - f(x_n)/f'(x_n)$. Near the root the number of correct digits roughly doubles each step (quadratic convergence, order 2), but it can fail from a poor start or where $f' \approx 0$. Applied to f', it finds minima.

Noise L4, L6

Measurement errors independent from sample to sample, with mean 0 and standard deviation σ. Not every unexplained variation is noise: *see* Time variability.

Noise floor L7

In PCA of noisy data, the level at which the variance of the components stops falling: independent noise adds the same variance σ_ε^2 to every direction, so the floor is flat; components above it carry signal. With correlated noise the floor is not flat, and where the signal ends is a judgement.

Nonautonomous system L2, L6, L7

A system with parameters or forcing that change explicitly in time. Its behaviour follows the time-varying parameters, and a time-independent steady state need not exist. Its phase space is the same as for the autonomous system, but the rule acting on it changes: the same state can lead to different futures, curves in phase space can cross, and with slowly changing parameters the fixed points and cycles of the frozen system move (*see* Frozen system). The change must be followed in time (time-resolved methods). Generally makes a series nonstationary; easily mistaken for noise.

Nonlinear mode decomposition (NMD) L7

Extracts oscillations from a time–frequency representation, follows their time-varying frequency and amplitude, keeps the harmonics of one oscillation together as one mode, and uses surrogates to decide oscillation or noise (Iatsenko, McClintock and Stefanovska, 2015). Suited to nonlinear, nonautonomous oscillators.

Nonparametric and parametric L5

A nonparametric method assumes no shape of the distribution and uses order, ranks, quantiles or resampling. A parametric method assumes a family (Gaussian, Poisson) and is right only if the family is right, which few values cannot check. Use a family only when prior physical knowledge supplies it.

Normal equations L3

$(A^\mathsf{T}A)\hat{\boldsymbol{\theta}} = A^\mathsf{T}\mathbf{b}$: the $p \times p$ linear system for a model linear in its parameters. Solve with QR or SVD (`np.linalg.lstsq`), not by forming $A^\mathsf{T}A$.

Null hypothesis L5

H_0: the statement whose consequences a test examines, e.g. both sets come from the same distribution, so any difference is due to chance.

Numerical integration L1, L4

Used in two senses. In L1: solving a differential equation step by step from its initial condition (time stepping). In L4: computing an integral $\int f\,dx$ from values of f (trapezoid, Simpson, quadrature).

Nyquist frequency L6

$f_\mathrm{N} = f_s/2 = 1/(2\Delta t)$: the highest frequency that samples taken every Δt can represent. *see* Aliasing.

O

Open system L2

Both mass and energy cross the boundary (e.g. the CSTR). Check: mass and energy balances.

Optimisation L3

Finding the x for which $f(x)$ is smallest: minima lie among the roots of ∇f. Methods: golden section, Nelder–Mead (f only), gradient descent, quasi-Newton BFGS (the general default), Newton (needs the Hessian).

Order of accuracy L2, L4

A method of order p has error $\approx Ch^p$, with C a constant: halving h divides the error by about 2^p. On a log–log plot the error is a straight line of slope p.

Overfitting L3

A model with too many parameters follows the noise: $\chi^2/\nu \ll 1$, and predictions outside the data fail.

P

Padding
L4, L6

Supplying values beyond the ends of the data by assumption (zeros, last value, mirror, periodic): a form of extrapolation, right only where its assumption holds. *Zero padding* before a DFT gives a finer grid of values, not a finer resolution.

Period
L5, L6

$P = 1/f$: the duration of one oscillation. Read in time, e.g. from upward zero crossings.

Permutation test
L5

Under H_0 the group labels do not matter: shuffle them many times and recompute the test statistic. p is the fraction of shuffles giving a value at least as large as the one observed.

Phase
L6

φ (rad): where in its cycle the oscillation is at $t = 0$. From the DFT, $\varphi_k = \arg X_k$.

Phase space
L7

The space of all possible states of a system, e.g. the plane $(x, \dot{x})$ for a second-order equation: the third way to present a time series, after the time and frequency domains. A rhythm becomes a closed curve, a resting state a point. Suited to autonomous dynamics; for nonautonomous systems it shows only a superposition of states from different times.

Poisson distribution
L3, L5

Counts of independent events: $p(k) = \mu^k e^{-\mu}/k!$, with standard deviation $\sqrt{\mu}$; for counts, $\sigma_i = \sqrt{N_i}$.

Population and sample
L1, L5

The population is the set of all values that could be measured; a sample is the finite set actually measured, from which population quantities are estimated.

Principal component analysis (PCA)
L7

Finds the straight directions along which data vary most: the SVD of the centred data matrix (equivalently, the eigenvectors of its covariance matrix). Principal directions $\mathbf{v}_k$, variance $\sigma_k^2/(N-1)$ along each, principal components $XV = U\Sigma$: each component is a fixed pattern over the variables ($\mathbf{v}_k$) times its profile over the rows. Uses only means, variances and covariances (linear statistics): data with the same covariances give the same PCA. Applies as it stands when the variation lies along straight directions that stay the same through the data (for a time series: a stationary system); it depends on units (*see* Standardising); components are uncorrelated, not independent causes. It ignores the order of the rows: shuffled rows give the same result, so for a time series the order enters only through delay coordinates. The same for independent measurements (rows: units) and for time series (rows: moments).

Pull
L3

$z = (\hat{\theta} - \theta_{\text{true}})/\sigma_{\hat{\theta}}$: the error of a fitted value in units of its own quoted uncertainty. Over many generated data sets, right error bars and no bias give mean 0 and standard deviation 1.

p-value
L5

The probability, if H_0 is true, of a test statistic at least as large as the one observed; $0 \le p \le 1$. A small p is evidence against H_0, or that a test assumption fails. A large p does not show that H_0 is true.

Q

Quantile
L5

The value below which a given fraction of the values lies; the median is the 50% quantile.

R

Random error
L3, L5

The part of a measurement error that changes from one measurement to the next. It averages out as $s/\sqrt{N}$ and is seen by statistics computed from the data.

Rank and rank test
L5

Rank: the position of a value in the sorted list. A rank test (Mann–Whitney) ranks the values of both groups together and asks whether one group's ranks are unusually low or high.

Rank correlation (Spearman)
L7

r_S: the correlation of the ranks (L5). Measures a monotonic relation; robust to bad values; the nonparametric choice.

Record length
L6

$T = N\Delta t$: the duration of the time series. It sets the frequency spacing and resolution, $\Delta f = 1/T$, and the lowest frequency that can be seen.

Reduced chi-square
L3

χ^2/ν. If the model and the σ_i are right, $\chi^2/\nu \approx 1 \pm \sqrt{2/\nu}$. Much larger: wrong model (often unmodelled time variability) or underestimated σ_i; much smaller: overestimated σ_i or overfitting.

Reference distribution
L5

How the test statistic would be distributed if H_0 were true. Nonparametric tests build it from the data (reshuffling, ranks); parametric tests derive it from an assumed distribution.

Residual
L2, L3

The difference between a data value and the fitted model, $r_i = y_i - f(x_i; \boldsymbol{\theta})$. Structured residuals show missing dynamics or time variability; only noise-like residuals may be described statistically.

Resolution (frequency)
L6

$\Delta f = 1/T$: two frequencies δf apart are seen as two only if the record is at least $T \gtrsim 1/\delta f$ long.

Richardson extrapolation
L4

If the error is Ch^p with p known, two results remove its leading part: $A \approx [2^p A(h/2) - A(h)]/(2^p - 1)$. Safe because it extrapolates in h, where the form of the error is known.

Root finding
L3

Finding x such that $f(x) = 0$: bisection, secant, Newton–Raphson, Brent. Bracket first (plot f), then let a fast method finish.

Round-off error
L1, L2, L4

The small error introduced every time a number is stored or an operation is performed, because of finite precision; relative size $\varepsilon \approx 10^{-16}$ (L4). Usually tiny, but it can accumulate, propagate, and be amplified by an ill-conditioned problem. It grows as h shrinks (e.g. ε/h for a first derivative), so the total error rises again below a best h.

Runge–Kutta methods L1, L2

Update rules that evaluate the slope at several points within each step and combine them. RK4: $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, with $k_1 = f(t_n, y_n)$, $k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$, $k_3 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$, $k_4 = f(t_n + h, y_n + hk_3)$. The family: RK2 (midpoint, order 2), RK3, RK4 (order 4, the default for smooth, non-stiff problems); RK45 is an adaptive pair.

Runge's example L4

A high-degree polynomial through equally spaced points oscillates near the ends, so more points can make the interpolation worse.

Running mean L6

The mean over a window that slides along the series; shows slow changes.

S

Sampling interval and rate L6

Δt: the time between samples; $f_s = 1/\Delta t$: samples per second. Δt sets the fastest oscillation that can be seen (*see* Nyquist frequency).

Secant method L3

Newton's method with the derivative replaced by the slope through the last two points; order ≈ 1.6, no f' needed.

Short-time Fourier transform L6

The windowed DFT of the data within a window of length L, slid along the record. Resolution about L in time and $1/L$ in frequency: one cannot have both.

Shuffle test L5, L6

Tests whether successive values are independent: compare the lag-1 autocorrelation r_1 of the data with that of many shuffled copies (surrogates).

Simpson's rule L4

Integration by a quadratic through each pair of intervals (N even): $\frac{h}{3}[f_0 + 4f_1 + 2f_2 + \cdots + 4f_{N-1} + f_N]$, order 4.

Singular spectrum analysis (SSA) L7

PCA of sliding windows (delay coordinates) of one series: each row of the data matrix is one window, e.g. $x_n, x_{n+4}, \ldots, x_{n+36}$. Assumes a stationary series.

Singular value decomposition (SVD) L7

Any $m \times n$ matrix $A = U\Sigma V^{\mathsf{T}}$: orthonormal (perpendicular, unit-length) input directions V, singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, orthonormal output directions U; $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ maps the unit circle to an ellipse. Gives the rank (number of nonzero σ_i), least squares (`lstsq`) and the closest matrix of rank k. For a data table, the $\mathbf{v}_i$ live in the space of the variables, the $\mathbf{u}_i$ in the space of the rows (units, or moments in time). Assumes nothing about the system; the assumptions belong to how its result is read.

Slope evaluation L1, L2

One computation of $f(t, y)$: the main cost of a time-stepping run. Compare methods at equal work, not per step.

Stability L1, L2

What happens to numerical errors as a calculation proceeds. A stable procedure keeps small errors under control; in an unstable one they grow until the calculation fails. Conditioning concerns the problem; convergence and stability concern the procedure. In L2: beyond a limit on h, a time-stepping solution becomes unstable.

Standard deviation L5

$s = \sqrt{\frac{1}{N-1}\sum(x_i - \bar{x})^2}$: the spread of single values.

Standard error of the mean L5

$s_{\bar{x}} = s/\sqrt{N}$: the standard deviation that $\bar{x}$ would have over repeated sets of N measurements. Valid only for independent values.

Standardising L7

$z_{ij} = (x_{ij} - \bar{x}_j)/s_j$: every column mean 0, standard deviation 1. Needed before PCA when variables have different units or scales (otherwise the units decide the directions); not for delay coordinates, one quantity in one unit. Sometimes loosely called normalising, which more often means scaling to $[0, 1]$ or to unit length.

State L7

The numbers that, with the equations, fix the future (for a deterministic system; with noise, only its probabilities), e.g. position and velocity $(x, \dot{x})$. For an autonomous system exactly one trajectory passes through each state.

Stationary series L6, L7

A series whose mean, standard deviation and autocorrelation do not change in time. A change of frequency or phase makes a series nonstationary even when mean and s stay the same.

Stiff problem L2

Processes act on very different time scales; an explicit method must take tiny steps to remain stable. Use an implicit or stiff solver.

Stochastic L2

Randomness is part of the dynamics: repeated runs from identical initial conditions give different trajectories. Distinct from *uncertainty* (inputs not known exactly) and from *time variability* (deterministic).

Superposition L2

The response to a sum of inputs is the sum of the responses: the defining property of a linear system.

Surrogate data L5, L6, L7

Copies of the data made to obey H_0 while keeping everything else, e.g. shuffled values: same values and histogram, order destroyed.

System boundary L2

The surface drawn around what we model; everything outside is the surroundings. What crosses it classifies the system as isolated, closed or open.

Systematic error L3, L5

The part of a measurement error that is the same in every measurement (e.g. a ruler 0.1% too long). It does not average out, is invisible to s, χ^2 and the bootstrap, and must be estimated separately by an independent check.

T

Test statistic L5

A number that measures how far apart two sets of values are, e.g. the difference of their medians.

Time series L6
Values of one quantity at successive times, kept in time order: $x_n = x(t_n)$, $t_n = n\Delta t$, $n = 0, \ldots, N-1$.

Time step L1, L2
h: the interval between successive values in a time-stepping method. It must resolve the fastest relevant time scale; too large and errors accumulate or the solution becomes unstable, too small and round-off grows.

Time variability L2, L3, L6
Variation caused by parameters or forcing that change in time: deterministic, not randomness, but easily mistaken for noise (a broad spectrum or a variance > 0 does not imply randomness).

Trajectory L7
The path of the state in phase space, starting from the initial condition. For an autonomous system trajectories never cross.

Trapezoid rule L4
Integration by straight lines between the points: $h[\frac{1}{2}f_0 + f_1 + \cdots + f_{N-1} + \frac{1}{2}f_N]$, order 2.

Truncation error L1, L4
The error from approximating an infinite or continuous process by a finite one, e.g. a finite step h, or the terms of a Taylor series left out. Kept distinct from round-off error. *see* Discretisation error.

U

Update rule L1, L2
A numerical method for $dy/dt = f(t, y)$ seen as a map from one point to the next, $(t_n, y_n) \to (t_{n+1}, y_{n+1})$, with $y_{n+1} \approx y(t_n + h)$, applied repeatedly. Euler and Runge–Kutta methods are update rules.

V

Validation L2, L3
Does the model represent the system? Comparison with experimental data and with independent observations; examining the assumptions.

Van der Pol oscillator L2, L3, L6, L7
$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$: negative damping for $|x| < 1$, positive for $|x| > 1$, giving a limit cycle. The module's example of a nonlinear system.

Variance L1, L5
The mean squared deviation from the mean. Population variance $\sigma^2 = \frac{1}{N}\sum(x_i - \bar{x})^2$; sample variance $s^2 = \frac{1}{N-1}\sum(x_i - \bar{x})^2$, the square of the standard deviation s.

Verification L2, L3
Did we solve the model correctly? Convergence tests, conserved quantities, exact solutions, and recovering known parameters from generated data.

W

Welford's algorithm L1
Computes the mean and variance incrementally: for each new value, $d_k = x_k - \bar{x}_{k-1}$, $\bar{x}_k = \bar{x}_{k-1} + d_k/k$, $M_{2,k} = M_{2,k-1} + d_k(x_k - \bar{x}_k)$; then $\sigma^2 = M_{2,N}/N$, $s^2 = M_{2,N}/(N-1)$. It works with quantities on the scale of the variation in the data, and so avoids catastrophic cancellation.

Window L6, L7
A function w_n that multiplies the data before the DFT, going smoothly to zero at both ends to reduce leakage, e.g. the Hann window. In the short-time Fourier transform, the segment of length L that slides along the record. In SSA (L7), one row of the data matrix: a stretch of the record that slides by one sample from row to row.

Symbols

Symbol	Meaning
h	step (time step; spacing of points)
p	order of a method (L2, L4); number of fitted parameters (L3); probability, p-value (L5)
C	constant in the error Ch^p (L2, L4); covariance matrix of parameters (L3) and of data (L7)
ε	tolerance (L3); relative precision $\approx 10^{-16}$ (L4); noise term (L2, L6, L7)
κ	condition number
J, H	Jacobian (L3, L7); Hessian (L3)
$\mathbf{x}^*$, $\boldsymbol{\delta}$	fixed point; deviation from it (L7)
X	data matrix, N rows $\times$ m variables (L7)
$\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}$	parameters, fitted parameters
r_i	residual (L3)
σ_i	standard uncertainty of a data point (L3); singular value (L7)
χ^2, ν	chi-square, degrees of freedom
ρ_{jk}	correlation of two parameters
z	pull (L3); standardised value (L7)
λ, $\mathbf{v}$	eigenvalue, eigenvector (L7)
U, Σ, V	factors of the SVD (L7)
s_{xy}	covariance (L7)
r, r_S	correlation, rank correlation (L7)

Symbol	Meaning
N	number of values
$\bar{x}$, s	mean, standard deviation
σ^2, s^2	population variance, sample variance (L1); σ_ε: standard deviation of the noise ε (L7)
$O(n)$	cost growing in proportion to n (L1)
$s_{\bar{x}}$	standard error of the mean
B	number of bootstrap repetitions
H_0	null hypothesis
r_1, r_k	lag-1 autocorrelation, autocorrelation at lag k
Δt, f_s	sampling interval, sampling rate
T	record length, $T = N\Delta t$
f, ω	frequency, angular frequency
P	period, $P = 1/f$
A, φ	amplitude, phase
X_k, f_k	DFT coefficient, its frequency k/T
Δf	frequency spacing and resolution, $1/T$
f_N	Nyquist frequency, $1/(2\Delta t)$
w_n	window
L	window length (short-time Fourier transform)
τ_c	correlation time
τ	lag (L6); delay (L7); decay time (L3)
N_0	initial count rate of the decay (L3)
$\mathcal{L}$	likelihood (L3)
μ	van der Pol parameter (L2, L6, L7)
x, $\dot{x}$	state of van der Pol: position, velocity (L7)